# ISAAC DOBIE

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## EDUCATION

**Massachusetts Institute of Technology** — Class of 2027
B.S. in Mechanical Engineering — GPA: 5.0/5.0 — *Cambridge, MA*

**Relevant Coursework:** Design and Manufacturing, Numerical Computation, Dynamics and Control, Mechanics of Materials, Electronics for Mechanical Systems, Thermal-Fluids Engineering

## EXPERIENCE

**General Motors** — June 2026 - Sep. 2026
*Active Thermal Hardware Intern* — *Warren, MI*

· Developing a full-stack AI application to analyze dealership service verbatims to identify root causes of thermal hardware warranty failures across vehicle platforms.

**MIT Rocket Team** — Sep. 2025 - Present
*Liquid Propulsion Subteam - Structures Member* — *Cambridge, MA*

· Designed structural components for a VTOL rocket prototype using SolidWorks, finite element analysis, and MATLAB calculations to validate strength and stiffness under launch loading.

**Cohen's Nonlinear Solid Mechanics Lab** — Feb. 2026 - Present
*Undergraduate Researcher* — *Cambridge, MA*

· Designed and prototyped a bench-top Volume-Controlled Cavity Expansion (VCCE) module for testing soft materials.
· Implemented a CoreXY positioning architecture enabling belt-driven 2-axis motion using stepper motor actuation.

**AutoStore** — June 2025 - Aug. 2025
*Solution Consulting Intern* — *Salem, NH*

· Built an optimization application to recommend robotic warehouse configurations using customer requirements while maximizing throughput and minimizing cost.

**MIT Mission Innovation X Lab** — Sep. 2024 - Jan. 2026
*Undergraduate Researcher* — *Cambridge, MA*

· Conducted fundamental research into software integration for flight test education and development for the U.S. Air Force.
· Developed correlation-based anomaly detection models for flight-test telemetry supporting U.S. Air Force research.

## PROJECTS

**Six-Axis Robotic Arm**

· Designing and building a six-axis robotic arm for automated 3D-printer operation.
· Integrating Raspberry Pi control, NEMA 17 stepper motors, and custom FDM 3D-printed components.

**2.007 Design Competition**

· Built an autonomous robot to compete in the 2.007 design competition using SolidWorks design, Arduino Nano Autonomy, and mill & lathe machining.
· Finished in the top 8 of more than 150 competitors; Awarded 'Hot Air Balloon Award' for an elegant and effective robot.

## SKILLS

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| **Programming:** | Python, R, C++, MATLAB |
| **Machining:** | CNC Mill, Lathe, Power Tools |
| **CAD/FEA:** | SolidWorks, Fusion 360 |

## LEADERSHIP & INVOLVEMENT

**MIT Varsity Basketball:** Team Captain, Rookie of the Year, Academic All-Conference — Sep. 2023 - Present
**Tau Beta Pi Engineering Honor Society:** Member — Mar. 2026 - Present
**Pi Tau Sigma Mechanical Engineering Honor Society:** Member — Dec. 2025 - Present
**Sigma Chi Fraternity:** Vice President, Risk Manager, House Manager — Sep. 2023 - Present